CSE STUDY HUB

Data Structures & Algorithms

UNIT 1 · Fundamentals, complexity & searching

What this unit is about

This unit builds understanding of **Fundamentals, complexity & searching**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Data structures and algorithms: definitions, characteristics and classifications

2. Abstract Data Types (ADT), operations and choosing a data structure

3. Time and space complexity; best, average and worst cases

4. Asymptotic notation: Big-O, Big-Omega and Big-Theta

5. Linear search: algorithm, trace and complexity

6. Binary search: iterative and recursive methods; sorted-input requirement

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 1 NOTES · PART 1

Data structures and algorithms: definitions, characteristics and classifications

Start with a precise definition of Data structures and algorithms: definitions, characteristics and classifications. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Abstract Data Types (ADT), operations and choosing a data structure

The OSI model has Physical, Data Link, Network, Transport, Session, Presentation and Application layers. Each layer provides services to the one above and uses the one below; encapsulation adds headers as data moves down.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Fundamentals, complexity & searching, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 2

Time and space complexity; best, average and worst cases

Arithmetic average = total ÷ number of observations. Weighted average = Σ(value×weight) ÷ Σ(weight). If one value changes by d, the total changes by d, which helps solve replacement questions.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Asymptotic notation: Big-O, Big-Omega and Big-Theta

Big-O gives an asymptotic upper bound on growth; Big-Ω gives a lower bound and Big-Θ gives a tight bound. Ignore constants and lower-order terms when describing large-input growth. Example: $3n^2+5n+2$ is $\Theta(n^2)$.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Fundamentals, complexity & searching, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 3

Linear search: algorithm, trace and complexity

Linear search examines items sequentially until the target is found or the collection ends. It works on unsorted data. Best case is O(1), worst case O(n), and it is useful for small or unsorted collections.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Binary search: iterative and recursive methods; sorted-input requirement

Binary search requires sorted data. Compare the target with the middle item and discard the half that cannot contain it. Repeating halves the search interval, giving O(log n) time; iterative search uses O(1) extra space.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Fundamentals, complexity & searching, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Data structures and algorithms: definitions, characteristics and classifications** — know the definition, core working and one example.

• **Abstract Data Types (ADT), operations and choosing a data structure** — know the definition, core working and one example.

• **Time and space complexity; best, average and worst cases** — know the definition, core working and one example.

• **Asymptotic notation: Big-O, Big-Omega and Big-Theta** — know the definition, core working and one example.

• **Linear search: algorithm, trace and complexity** — know the definition, core working and one example.

• **Binary search: iterative and recursive methods; sorted-input requirement** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Fundamentals, complexity & searching. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Data structures and algorithms: definitions, characteristics and classifications and explain its importance.

Q2. Explain the working/steps involved in Abstract Data Types (ADT), operations and choosing a data structure.

Q3. Differentiate between two related concepts from Fundamentals, complexity & searching.

Q4. Give one practical application of Binary search: iterative and recursive methods; sorted-input requirement and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Fundamentals, complexity & searching.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.